

Multidimensional BSDEs with Weak Monotonicity and General Growth Generators

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Abstract This paper aims at solving a multidimensional backward stochastic differential equation (BSDE) whose generator g satisfies a weak monotonicity condition and a general growth condition in y . We first establish an existence and uniqueness result of solutions for this kind of BSDEs by using systematically the technique of the priori estimation, the convolution approach, the iteration, the truncation and the Bihari inequality. Then, we overview some assumptions related closely to the monotonicity condition in the literature and compare them in an effective way, which yields that our existence and uniqueness result really and truly unifies the Mao condition in y and the monotonicity condition with the general growth condition in y , and it generalizes some known results. Finally, we prove a stability theorem and a comparison theorem for this kind of BSDEs, which also improves some known results.

Keywords Backward stochastic differential equation, existence and uniqueness, weak monotonicity condition, general growth condition, comparison theorem

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1 Introduction

Suppose that k and d are two given positive integers. Let $(\Omega, \mathcal{F}, P)$ be a complete probability space carrying a standard d -dimensional Brownian motion $(B_t)_{t \geq 0}$. Let $(\mathcal{F}_t)_{t \geq 0}$ be the natural σ -algebra generated by $(B_t)_{t \geq 0}$ augmented by the P -null sets of $\mathcal{F}$. All the measurability notions will refer to this filtration. Let $T > 0$ be a given real number. In this paper, we always work in the space $(\Omega, \mathcal{F}_T, P)$, and only consider processes indexed by $t \in [0, T]$. The Euclidean norm of a vector $y \in \mathbb{R}^k$ will be defined by $|y|$, and for a $k \times d$ matrix z , we define $|z| = \sqrt{\text{Tr}(zz^*)}$, where z^* is the transpose of z . Let $\langle x, y \rangle$ represent the inner product of $x, y \in \mathbb{R}^k$. We denote by $L^2(\mathcal{F}_T; \mathbb{R}^k)$ the set of all $\mathbb{R}^k$ -valued, square integral and $\mathcal{F}_T$ -measurable random vectors.

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Let $\mathcal{S}^2(0, T; \mathbb{R}^k)$ denote the set of $\mathbb{R}^k$ -valued, adapted and continuous processes $(\phi_t)_{t \in [0, T]}$ such that

$$\|\phi\|_{\mathcal{S}^2}^2 := \mathbf{E} \left[\sup_{t \in [0, T]} |\phi_t|^2 \right] < +\infty.$$

Moreover, let $M^2(0, T; \mathbb{R}^k)$ (resp. $M^2(0, T; \mathbb{R}^{k \times d})$) denote the set of $(\mathcal{F}_t)$ -progressively measurable $\mathbb{R}^k$ -valued ($\mathbb{R}^{k \times d}$ -valued) processes $(\varphi_t)_{t \in [0, T]}$ such that

$$\|\varphi\|_{M^2}^2 := \mathbf{E} \left[\int_0^T |\varphi_t|^2 dt \right] < +\infty.$$

Obviously, $\mathcal{S}^2(0, T; \mathbb{R}^k)$ is a Banach space, and both $M^2(0, T; \mathbb{R}^k)$ and $M^2(0, T; \mathbb{R}^{k \times d})$ are Hilbert spaces.

In this paper, we consider the following multidimensional backward stochastic differential equation (BSDE for short in the remaining):

$$y_t = \xi + \int_t^T g(s, y_s, z_s) ds - \int_t^T z_s dB_s, \quad t \in [0, T], \quad (1.1)$$

where T is called the time horizon, $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^k)$ is called the terminal condition, the random function $g(\omega, t, y, z) : \Omega \times [0, T] \times \mathbb{R}^k \times \mathbb{R}^{k \times d} \rightarrow \mathbb{R}^k$ is $(\mathcal{F}_t)$ -progressively measurable for each (y, z) , called the generator of BSDE (1.1). The triple (ξ, T, g) is called the coefficients (parameters) of BSDE (1.1).

Definition 1.1 A pair of processes $(y_t, z_t)_{t \in [0, T]}$ is called a solution to BSDE (1.1), if $(y_t, z_t)_{t \in [0, T]} \in \mathcal{S}^2(0, T; \mathbb{R}^k) \times M^2(0, T; \mathbb{R}^{k \times d})$ and satisfies (1.1).

BSDEs, in a nonlinear form, were initially introduced by Pardoux and Peng [16] who established an existence and uniqueness result of solutions under the Lipschitz assumption of the generator g . Since then, much effort has been made to weaken the Lipschitz hypothesis on g , for instance, in the one-dimensional setting ($k = 1$), Lepeltier and San Martin [14] proved the existence of a solution for BSDE (1.1) when g is continuous and of linear growth in (y, z) , and Fan et al. [7] extended it to the case where the time horizon T may be infinite. Kobylanski [12] obtained the existence and uniqueness of a solution when g has a quadratic growth in z and the terminal condition ξ is bounded, and then [2, 3] further extended the result of [12] to the case of unbounded terminal conditions. In the multidimensional case, Hamadène [11] and Fan et al. [8] investigated the existence and uniqueness of a solution for (1.1) where g is uniformly continuous in (y, z) and for each $i = 1, 2, \dots, k$, the i -th component g_i of g depends only on the i -th row of the matrix z . Mao [15] obtained an existence and uniqueness result of a solution for (1.1), where g satisfies a particular non-Lipschitz condition in y called usually the Mao condition in the literature, and [9, 10] extended it to the case where the time horizon T may be infinite. Peng [19] first introduced a monotonicity condition in y for g , and under this monotonicity condition as well as a general growth condition in y for g , Pardoux [17] established an existence and uniqueness result of a solution for (1.1). Using the same monotonicity condition and more general growth condition in y for g , Briand et al. [4] investigated the existence and uniqueness of a solution in L^p ($p > 1$) for (1.1). Furthermore, Situ [20] put forward a weak monotonicity condition as well as a linear growth condition in y for g and considered the existence and uniqueness of solutions to BSDEs with jumps. The reader is referred to [18, 21–24] for some

further results on BSDEs with jumps. We should point out that all of these results need the linear growth condition in y for g .

This paper is interested in solving a multidimensional BSDE whose generator g satisfies the weak monotonicity condition and the general growth condition in y mentioned above. We first establish an existence and uniqueness result of solutions for this kind of BSDEs (see Theorem 2.1 in Section 2), which is one of the main results of this paper. It should be mentioned that in the procedure of the proof of Theorem 2.1, some new difficulties arise due to the combination of the weak monotonicity condition and the general growth condition in y for g , particularly, the weak convergence technique used in [17] cannot work in our setting. These difficulties are all overcome by using systematically the technique of the a priori estimation, the convolution approach, the iteration, the truncation and the Bihari inequality. Then, we overview some assumptions related closely to the monotonicity condition in the literature and compare them in an effective way, which yields that Theorem 2.1 really and truly unifies the Mao condition in y and the monotonicity condition with the general growth condition in y , and it generalizes some corresponding results in [6, 11, 15–17]. Finally, we prove a stability theorem and a comparison theorem for this kind of BSDEs, which also improves the corresponding comparison results in [5] and [17].

This paper is organized as follows. In Section 2, we put forward the existence and uniqueness result and introduce some remarks and examples to show that it generalizes the corresponding results mentioned above. In Section 3, we start with a priori estimate and then prove the existence and uniqueness result. Then, we establish a stability theorem in Section 4 and a comparison theorem in Section 5. Finally, some technical details in Section 2, including the proof of the relations of the assumptions related to the monotonicity condition, are provided in Appendix.

2 An Existence and Uniqueness Result

In this section, we will put forward the existence and uniqueness result and introduce some remarks and examples to show that it generalizes the corresponding results in [6, 11, 15–17]. Let us start with introducing the following assumptions:

(H1) g satisfies the weak monotonicity condition in y , i.e., there exists a nondecreasing and concave function $\kappa(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\kappa(0) = 0$, $\kappa(u) > 0$ for $u > 0$ and $\int_{0+} \kappa^{-1}(u) du = +\infty$ such that $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\langle y_1 - y_2, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \rangle \leq \kappa(|y_1 - y_2|^2).$$

(H2) $dP \times dt$ -a.e., $\forall z \in \mathbb{R}^{k \times d}$, $y \mapsto g(\omega, t, y, z)$ is continuous.

(H3) g has a general growth with respect to y , i.e., $dP \times dt$ -a.e.,

$$\forall y \in \mathbb{R}^k, \quad |g(\omega, t, y, 0)| \leq |g(\omega, t, 0, 0)| + \varphi(|y|),$$

where $\varphi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is an increasing continuous function.

(H4) g is Lipschitz continuous in z , uniformly with respect to (ω, t, y) , i.e., there exists a constant $\mu \geq 0$ such that $dP \times dt$ -a.e.,

$$\forall y \in \mathbb{R}^k, \quad z_1, z_2 \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq \mu |z_1 - z_2|.$$

(H5) $g(\omega, t, 0, 0) \in M^2(0, T; \mathbb{R}^k)$.

The following Theorem 2.1 is one of the main results of this paper.

Theorem 2.1 *Let g satisfy assumptions (H1)–(H5). Then, for each $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^k)$, the BSDE with parameters (ξ, T, g) has a unique solution.*

In order to compare Theorem 2.1 with some known results, let us further introduce the following assumptions on g :

(H1a) g satisfies the one-sided Osgood condition in y , i.e., there exists a nondecreasing and concave function $\rho(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\rho(0) = 0$, $\rho(u) > 0$ for $u > 0$ and $\int_{0+} \rho^{-1}(u) du = +\infty$ such that $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq \rho(|y_1 - y_2|).$$

(H1b) g satisfies the one-sided Mao condition in y , i.e., there exists a nondecreasing and concave function $\kappa(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\kappa(0) = 0$, $\kappa(u) > 0$ for $u > 0$ and $\int_{0+} \kappa^{-1}(u) du = +\infty$ such that $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq \sqrt{\kappa(|y_1 - y_2|^2)}.$$

(H1c) g satisfies the one-sided Constantin condition in y , i.e., there exists a nondecreasing and concave function $\rho(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\rho(0) = 0$, $\rho(u) > 0$ for $u > 0$ and $\int_{0+} u \rho^{-2}(u) du = +\infty$ such that $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq \rho(|y_1 - y_2|).$$

(H1d) g satisfies the monotonicity condition in y , i.e., there exists a constant $\lambda \geq 0$ such that $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq \lambda |y_1 - y_2|.$$

(H1)' There exists a nondecreasing and concave function $\rho(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\rho(0) = 0$, $\rho(u) > 0$ for $u > 0$ and $\int_{0+} \rho^{-1}(u) du = +\infty$ such that $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |y_1 - y_2| |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq \rho(|y_1 - y_2|^2).$$

(H1a)' g satisfies the Osgood condition in y , i.e., there exists a nondecreasing and concave function $\rho(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\rho(0) = 0$, $\rho(u) > 0$ for $u > 0$ and $\int_{0+} \rho^{-1}(u) du = +\infty$ such that $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq \rho(|y_1 - y_2|).$$

(H1b)' g satisfies the Mao condition in y , i.e., there exists a nondecreasing and concave function $\kappa(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\kappa(0) = 0$, $\kappa(u) > 0$ for $u > 0$ and $\int_{0+} \kappa^{-1}(u) du = +\infty$ such that $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)|^2 \leq \kappa(|y_1 - y_2|^2).$$

(H1c)' g satisfies the Constantin condition in y , i.e., there exists a nondecreasing and concave function $\rho(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\rho(0) = 0$, $\rho(u) > 0$ for $u > 0$ and $\int_{0+} u \rho^{-2}(u) du = +\infty$

such that $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq \rho(|y_1 - y_2|).$$

(H1d)' g is Lipschitz continuous in y , uniformly with respect to (ω, t, z) , i.e., there exists a constant $\lambda \geq 0$ such that $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq \lambda |y_1 - y_2|.$$

(H6) For each $i = 1, 2, \dots, k$, the i -th component g_i of g depends only on the i -th row of the matrix z .

Remark 2.2 Clearly, (H1)' (resp. (H1a)', (H1b)', (H1c)', (H1d)') implies (H1) (resp. (H1a), (H1b), (H1c), (H1d)), (H1d) (resp. (H1d)') implies (H1c) (resp. (H1c)'), and (H1a)' implies (H2)+(H3). In Appendix, we will show that (H1c) $\Rightarrow$ (H1b) $\Rightarrow$ (H1a) $\Rightarrow$ (H1) and (H1c)' $\Rightarrow$ (H1b)' $\Rightarrow$ (H1a)' $\Rightarrow$ (H1)'.

In [11] (resp. [6, 15–17]), the authors established the existence and uniqueness result for BSDE (1.1) under assumptions (H1a)' and (H4)–(H6) (resp. (H1c)' and (H4)–(H5), (H1d) and (H2)–(H5), (H1b)' and (H4)–(H5), (H1d)' and (H4)–(H5)). Therefore, it follows from Remark 2.2 that Theorem 2.1 generalizes these results. Furthermore, by Remark 2.2 and Theorem 2.1, we also know that the existence and uniqueness result for BSDE (1.1) holds also true under assumptions (H1)' and (H2)–(H5) (resp. (H1a) and (H2)–(H5), (H1b) and (H2)–(H5), (H1c) and (H2)–(H5), (H1a)' and (H4)–(H5)), none of which is known in existing literature.

Remark 2.3 It is clear that (H1c) (resp. (H1c)') is strictly weaker than (H1d) (resp. (H1d)'). The following Example 2.4 shows that (H1a) is strictly weaker than (H1b), and (H1) is also strictly weaker than (H1)'. Similarly, we also know that (H1a)' is strictly weaker than (H1b)'. However, so far we still do not know that whether (H1) (resp. (H1)') is indeed strictly weaker than (H1a) (resp. (H1a)'), and whether (H1b) (resp. (H1b)') is strictly weaker than (H1c) (resp. (H1c)').

Example 2.4 Let $k = 1$ and $g(t, y, z) = e^{-y} + h(|y|) + \sin |z| + |B_t|$, where

$$h(x) = \begin{cases} -x \ln x, & 0 < x \leq \delta, \\ h'(\delta-)(x - \delta) + h(\delta), & x > \delta, \\ 0, & \text{otherwise,} \end{cases}$$

with $\delta > 0$ small enough.

In Appendix, it is shown that g satisfies neither (H1b) nor (H1)'. Then none of the known results can answer the question on the existence and uniqueness of solutions for the BSDE with this generator g . However, it is clear that g satisfies (H2)–(H5) by choosing $\varphi(y) = e^y + h(y)$ and $\mu = 1$. And, we can prove that g satisfies (H1a) and then (H1) by verifying that e^{-y} is a decreasing function in y , $\int_0^+ h^{-1}(u)du = +\infty$ and $h(\cdot)$ is concave and sub-additive on $\mathbb{R}^+$, then the following inequality holds

$$\forall y_1, y_2, z, \quad \left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq h(|y_1 - y_2|).$$

Thus, this generator g satisfies all conditions in Theorem 2.1. Hence, the BSDE with parameters (ξ, T, g) has a unique solution for each $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^k)$.

The following is an example of a multidimensional BSDE whose generator g satisfies (H1)–(H5), but neither (H6), (H1b) nor (H1)'.

Example 2.5 Let $k = 2$ and for each $y = (y_1, y_2) \in \mathbb{R}^2$ and $z \in \mathbb{R}^{2 \times d}$, let $g(t, y, z) = (g_1(t, y, z), g_2(t, y, z))$ be defined by

$$g_1(t, y, z) = e^{-y_1} + h(|y_1| + |y_2|) \quad \text{and} \quad g_2(t, y, z) = e^{-y_2} + |z| + |B_t|,$$

where $h(x)$ is defined in Example 2.4.

Obviously, this generator g does not satisfy (H6). Furthermore, we can also prove that g satisfies neither (H1b) nor (H1)'. However, it is not difficult to check that g satisfies (H1a) with $\rho(u) = h(u)$, (H2), (H3) with $\varphi(y) = e^y + h(y)$, (H4) with $\mu = 1$ and (H5). Hence, it follows from Theorem 2.1 that the BSDE with parameters (ξ, T, g) has a unique solution for each $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^k)$.

Remark 2.6 In Appendix, it is shown that we can use the continuity instead of the concavity of $\rho(\cdot)$ and $\kappa(\cdot)$ in (H1a)', (H1c)' and (H1b)'.

3 Proof of the Existence and Uniqueness Result

In this section, we will give the proof of Theorem 2.1 in Section 2 by using systematically the technique mentioned in Introduction, some of which are borrowed, respectively, from [4, 15, 17, 20, 23, 24]. It should be particularly mentioned that the weak convergence technique used in [17] cannot work in our setting. Let us first introduce an a priori estimate, which will play an important role in the proof of Theorem 2.1. In this estimate, we will use the following assumption on g :

$$(A) \quad dP \times dt\text{-a.e.}, \forall (y, z) \in \mathbb{R}^k \times \mathbb{R}^{k \times d}, \langle y, g(\omega, t, y, z) \rangle \leq \psi(|y|^2) + \lambda|y||z| + |y|\varphi_t,$$

where λ is a positive constant, φ_t is a non-negative and $(\mathcal{F}_t)$ -progressively measurable processes with $\mathbf{E}[\int_0^T \varphi_t^2 dt] < +\infty$, and $\psi(\cdot)$ is a nondecreasing concave function from $\mathbb{R}^+$ to itself with $\psi(0) = 0$.

Proposition 3.1 Let g satisfy (A) and let $(y_t, z_t)_{t \in [0, T]}$ be a solution to the BSDE with parameters (ξ, T, g) . Then, for each $\theta > 0$, there exists a constant $c > 0$ depending only on λ and θ such that, for each $0 \leq u \leq t \leq T$,

$$\begin{aligned} & \mathbf{E} \left[\sup_{r \in [t, T]} |y_r|^2 \middle| \mathcal{F}_u \right] + \mathbf{E} \left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u \right] \\ & \leq e^{c(T-t)} \left\{ c \mathbf{E} [|\xi|^2 | \mathcal{F}_u] + c \int_t^T \psi(\mathbf{E} [|y_s|^2 | \mathcal{F}_u]) ds + \frac{1}{\theta} \mathbf{E} \left[\int_t^T \varphi_s^2 ds \middle| \mathcal{F}_u \right] \right\}. \end{aligned} \quad (3.1)$$

Proof Applying Itô's formula to $|y_t|^2$ yields that, for each $t \in [0, T]$,

$$|y_t|^2 + \int_t^T |z_s|^2 ds = |\xi|^2 + 2 \int_t^T \langle y_s, g(s, y_s, z_s) \rangle ds - 2 \int_t^T \langle y_s, z_s dB_s \rangle. \quad (3.2)$$

By the assumption (A) and the inequality $2ab \leq \theta a^2 + b^2/\theta$ for each $\theta > 0$, we have

$$2 \langle y_s, g(s, y_s, z_s) \rangle \leq 2\psi(|y_s|^2) + 2\lambda|y_s||z_s| + 2|y_s|\varphi_s$$

$$\leq 2\psi(|y_s|^2) + (2\lambda^2 + 290\theta)|y_s|^2 + \frac{\varphi_s^2}{290\theta} + \frac{1}{2}|z_s|^2. \quad (3.3)$$

Thus, it follows from (3.2) and (3.3) that, for each $0 \leq u \leq t \leq T$,

$$\frac{1}{2}\mathbf{E}\left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u\right] \leq X_u^t, \quad (3.4)$$

where

$$\begin{aligned} X_u^t &= \mathbf{E}[|\xi|^2 | \mathcal{F}_u] + (2\lambda^2 + 290\theta) \int_t^T \mathbf{E}\left[\sup_{r \in [s, T]} |y_r|^2 \middle| \mathcal{F}_u\right] ds \\ &\quad + \mathbf{E}\left[\int_t^T (2\psi(|y_s|^2) + \frac{\varphi_s^2}{290\theta}) ds \middle| \mathcal{F}_u\right]. \end{aligned}$$

Furthermore, it follows from the Burkholder–Davis–Gundy inequality that $\{M_t := \int_0^t \langle y_s, z_s dB_s \rangle\}_{t \in [0, T]}$ is a uniformly integrable martingale. In fact, for each $0 \leq u \leq t \leq T$, we have

$$\begin{aligned} 2\mathbf{E}\left[\sup_{r \in [t, T]} \left|\int_r^T \langle y_s, z_s dB_s \rangle\right| \middle| \mathcal{F}_u\right] &\leq 12\mathbf{E}\left[\sup_{r \in [t, T]} |y_r| \cdot \left(\int_t^T |z_s|^2 ds\right)^{1/2} \middle| \mathcal{F}_u\right] \\ &\leq \frac{1}{2}\mathbf{E}\left[\sup_{r \in [t, T]} |y_r|^2 \middle| \mathcal{F}_u\right] + 72\mathbf{E}\left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u\right] \\ &< +\infty. \end{aligned} \quad (3.5)$$

By virtue of (3.3) and (3.5), it follows from (3.2) that

$$\mathbf{E}\left[\sup_{r \in [t, T]} |y_r|^2 \middle| \mathcal{F}_u\right] + \mathbf{E}\left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u\right] \leq 2X_u^t + 144\mathbf{E}\left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u\right].$$

Thus, combining the above inequality and (3.4) yields that

$$\mathbf{E}\left[\sup_{r \in [t, T]} |y_r|^2 \middle| \mathcal{F}_u\right] + \mathbf{E}\left[\int_t^T |z_s|^2 ds \middle| \mathcal{F}_u\right] \leq 290X_u^t,$$

and then, by letting $h(t) = \mathbf{E}[\sup_{r \in [t, T]} |y_r|^2 | \mathcal{F}_u] + \mathbf{E}[\int_t^T |z_s|^2 ds | \mathcal{F}_u]$ and noticing the definition of X_u^t , we know that, for each $0 \leq u \leq t \leq T$,

$$\begin{aligned} h(t) &\leq 290\mathbf{E}[|\xi|^2 | \mathcal{F}_u] + 290(2\lambda^2 + 290\theta) \int_t^T h(s) ds \\ &\quad + \mathbf{E}\left[\int_t^T \left(580\psi(|y_s|^2) + \frac{\varphi_s^2}{\theta}\right) ds \middle| \mathcal{F}_u\right], \end{aligned}$$

from which together with Gronwall's inequality, Fubini's theorem and Jensen's inequality, in view of the concavity condition of $\psi(\cdot)$, the inequality (3.1) follows. The proof is completed. $\square$

Now, we can give the proof of Theorem 2.1.

Proof of Theorem 2.1 Suppose that g satisfies the assumptions (H1)–(H5). Let us start with studying the uniqueness part. Let $(y_t^i, z_t^i)_{t \in [0, T]}$ ($i = 1, 2$) be two solutions of the BSDE with parameters (ξ, T, g) . Then $(y_t^1 - y_t^2, z_t^1 - z_t^2)_{t \in [0, T]}$ is a solution to the following BSDE:

$$y_t = \int_t^T \hat{g}(s, y_s, z_s) ds - \int_t^T z_s dB_s, \quad t \in [0, T], \quad (3.6)$$

where, for each $(y, z) \in \mathbb{R}^k \times \mathbb{R}^{k \times d}$, $\hat{g}(s, y, z) := g(s, y + y_s^2, z + z_s^2) - g(s, y_s^2, z_s^2)$. It follows from (H1) and (H4) that $dP \times dt$ -a.e.,

$$\forall (y, z) \in \mathbb{R}^k \times \mathbb{R}^{k \times d}, \quad \langle y, \hat{g}(t, y, z) \rangle \leq \kappa(|y|^2) + \mu|y||z|.$$

Then the assumption (A) is satisfied for the generator $\hat{g}(t, y, z)$ of BSDE (3.6) with $\psi(u) = \kappa(u)$, $\lambda = \mu$ and $\varphi_t \equiv 0$. Thus, it follows from Proposition 3.1 with $u = 0$ and $\theta = 1$ that there exists a constant $c > 0$ depending only on μ and T such that, for $t \in [0, T]$,

$$\mathbf{E} \left[\sup_{r \in [t, T]} |y_r^1 - y_r^2|^2 \right] + \mathbf{E} \left[\int_t^T |z_s^1 - z_s^2|^2 ds \right] \leq c \int_t^T \kappa \left(\mathbf{E} \left[\sup_{r \in [s, T]} |y_r^1 - y_r^2|^2 \right] \right) ds.$$

In view of $\int_{0+} \kappa^{-1}(u) du = +\infty$, Bihari's inequality (see [1]) yields that, for each $t \in [0, T]$,

$$\mathbf{E} \left[\sup_{r \in [t, T]} |y_r^1 - y_r^2|^2 \right] + \mathbf{E} \left[\int_t^T |z_s^1 - z_s^2|^2 ds \right] = 0.$$

The proof of the uniqueness part of Theorem 2.1 is then complete.

Let us turn to the existence part. The proof will be split into four steps.

Step 1 We prove that for each $V \in M^2(0, T; \mathbb{R}^{k \times d})$ and $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^k)$, if there exists a positive constant K such that

$$dP\text{-a.s.}, |\xi| \leq K, \quad dP \times dt\text{-a.e.}, |g(t, 0, 0)| \leq K \quad \text{and} \quad |V_t| \leq K, \quad (3.7)$$

then there exists a unique solution to the following BSDE:

$$y_t = \xi + \int_t^T g(s, y_s, V_s) ds - \int_t^T z_s dB_s, \quad t \in [0, T]. \quad (3.8)$$

Let ψ be a function of $C^\infty(\mathbb{R}^k, \mathbb{R}^+)$ with the closed unit ball as compact support, and satisfy $\int_{\mathbb{R}^k} \psi(y) dy = 1$. For each $n \geq 1$ and each $(\omega, t, y) \in \Omega \times [0, T] \times \mathbb{R}^k$, we set

$$g_n(t, y, V_t) := n^k g(t, y, V_t) * \psi(ny) = n^k \int_{\mathbb{R}^k} g(t, u, V_t) \psi(ny - nu) du. \quad (3.9)$$

Then, g_n is an $(\mathcal{F}_t)$ -progressively measurable process for each $y \in \mathbb{R}^k$ and

$$g_n(t, y, V_t) = \int_{\mathbb{R}^k} g \left(t, y - \frac{u}{n}, V_t \right) \psi(u) du = \int_{\{u: |u| \leq 1\}} g \left(t, y - \frac{u}{n}, V_t \right) \psi(u) du. \quad (3.10)$$

It follows from (H3), (H4) and (3.7) that, for each $y \in \mathbb{R}^k$, $dP \times dt$ -a.e.,

$$|g(t, y, V_t)| \leq |g(t, y, 0)| + |g(t, y, V_t) - g(t, y, 0)| \leq K + \varphi(|y|) + \mu K. \quad (3.11)$$

Thus, it is easily checked from (3.9) that for each $n \geq 1$, $g_n(t, y, V_t)$ is locally Lipschitz in y , uniformly with respect to (t, ω) . Furthermore, for each $n \geq 1$ and $y \in \mathbb{R}^k$, it follows from (3.10) and (3.11) that $dP \times dt$ -a.e.,

$$|g_n(t, y, V_t)| \leq K + \varphi(|y| + 1) + \mu K. \quad (3.12)$$

Now, for some large enough integer $r > 0$ which will be chosen later, let θ_r be a smooth function such that $0 \leq \theta_r \leq 1$, $\theta_r(y) = 1$ for $|y| \leq r$ and $\theta_r(y) = 0$ as soon as $|y| \geq r + 1$. Then for each $n \geq 1$, the function $\theta_r(y) g_n(t, y, V_t)$ is globally Lipschitz in y , uniformly with respect to (t, ω) . Indeed, let us pick y and y' in $\mathbb{R}^k$. If $|y| > r + 1$ and $|y'| > r + 1$, then the statement is trivially satisfied and thus we reduce to the case $|y'| \leq r + 1$. Note that $g_n(t, y, V_t)$ is locally Lipschitz

in y , uniformly with respect to (t, ω) , and θ_r is globally Lipschitz in y . It follows from (3.12) that there exist two positive constants C_1 and C_2 such that $dP \times dt$ -a.e.,

$$\begin{aligned} & |\theta_r(y)g_n(t, y, V_t) - \theta_r(y')g_n(t, y', V_t)| \\ & \leq |\theta_r(y)||g_n(t, y, V_t) - g_n(t, y', V_t)| + |\theta_r(y) - \theta_r(y')||g_n(t, y', V_t)| \\ & \leq C_1|y - y'| + C_2(K + \mu K + \varphi(r + 2))|y - y'|. \end{aligned}$$

Thus, from [16], we know that for each $n \geq 1$, the following BSDE has a unique solution $(y_t^n, z_t^n)_{t \in [0, T]}$:

$$y_t^n = \xi + \int_t^T \theta_r(y_s^n)g_n(s, y_s^n, V_s)ds - \int_t^T z_s^n dB_s, \quad t \in [0, T]. \quad (3.13)$$

It follows from (H1) and (3.10) that, for each $n \geq 1$ and $y_1, y_2 \in \mathbb{R}^k$, $dP \times dt$ -a.e.,

$$\begin{aligned} & \langle y_1 - y_2, g_n(t, y_1, V_t) - g_n(t, y_2, V_t) \rangle \\ & \leq \int_{\mathbb{R}^k} \left\langle y_1 - y_2, g\left(t, y_1 - \frac{u}{n}, V_t\right) - g\left(t, y_2 - \frac{u}{n}, V_t\right) \right\rangle \psi(u) du \\ & \leq \int_{\mathbb{R}^k} \kappa(|y_1 - y_2|^2) \psi(u) du = \kappa(|y_1 - y_2|^2). \end{aligned} \quad (3.14)$$

For each $n \geq 1$ and $y \in \mathbb{R}^k$, combining (3.14) and (3.12) yields that $dP \times dt$ -a.e.,

$$\begin{aligned} \langle y, \theta_r(y)g_n(t, y, V_t) \rangle &= \theta_r(y) \langle y, g_n(t, y, V_t) - g_n(t, 0, V_t) \rangle + \theta_r(y) \langle y, g_n(t, 0, V_t) \rangle \\ &\leq \kappa(|y|^2) + |y|(K + \mu K + \varphi(1)). \end{aligned}$$

Then the assumption (A) is satisfied for the generator $\theta_r(y)g_n(t, y, V_t)$ of BSDE (3.13) with $\psi(u) = \kappa(u)$, $\lambda = 0$ and $\varphi_t \equiv K + \mu K + \varphi(1)$. Thus, in view of (3.7), it follows from Proposition 3.1 with $\theta = 1$ that there exists a constant $c > 0$ depending only on T such that, for each $n \geq 1$ and each $0 \leq u \leq t \leq T$,

$$\begin{aligned} & \mathbf{E}[|y_t^n|^2 | \mathcal{F}_u] + \mathbf{E}\left[\int_t^T |z_s^n|^2 ds \middle| \mathcal{F}_u\right] \\ & \leq cK^2 + c \int_t^T \kappa(\mathbf{E}[|y_s^n|^2 | \mathcal{F}_u])ds + c(K + \mu K + \varphi(1))^2 T. \end{aligned}$$

Furthermore, since $\kappa(\cdot)$ is a nondecreasing and concave function with $\kappa(0) = 0$, it increases at most linearly, i.e., there exists a constant $A > 0$ such that $\kappa(x) \leq A(x + 1)$ for each $x \geq 0$. Using Gronwall's inequality yields that

$$\begin{aligned} & \mathbf{E}[|y_t^n|^2 | \mathcal{F}_u] + \mathbf{E}\left[\int_t^T |z_s^n|^2 ds \middle| \mathcal{F}_u\right] \\ & \leq (cK^2 + cAT + c(K + \mu K + \varphi(1))^2 T) \cdot e^{cAT} := r^2. \end{aligned}$$

Substituting $u = t$ in the previous inequality yields that, for each $n \geq 1$,

$$\forall t \in [0, T], \quad |y_t^n| \leq r \quad \text{and} \quad \mathbf{E}\left[\int_0^T |z_s^n|^2 ds\right] \leq r^2. \quad (3.15)$$

By (3.13) and (3.15), we can conclude that $(y_t^n, z_t^n)_{t \in [0, T]}$ solves the following BSDE:

$$y_t^n = \xi + \int_t^T g_n(s, y_s^n, V_s)ds - \int_t^T z_s^n dB_s, \quad t \in [0, T]. \quad (3.16)$$

In the sequel, we shall show that $\{(y_t^n, z_t^n)_{t \in [0, T]}\}_{n=1}^\infty$ is a Cauchy sequence in the process space $\mathcal{S}^2(0, T; \mathbb{R}^k) \times M^2(0, T; \mathbb{R}^{k \times d})$. In fact, for each $n \geq 1$ and $m \geq 1$, let $\hat{y}^{n,m} = y_t^n - y_t^m$, $\hat{z}^{n,m} = z_t^n - z_t^m$. Then

$$\hat{y}_t^{n,m} = \int_t^T \hat{g}^{n,m}(s, \hat{y}_s^{n,m}, V_s) ds - \int_t^T \hat{z}_s^{n,m} dB_s, \quad t \in [0, T], \quad (3.17)$$

where $\hat{g}^{n,m}(s, y, V_s) := g_n(s, y + y_s^m, V_s) - g_m(s, y_s^m, V_s)$ for each $y \in \mathbb{R}^k$. It follows from (3.14) that for each $y \in \mathbb{R}^k$, $dP \times dt$ -a.e.,

$$\begin{aligned} \langle y, \hat{g}^{n,m}(t, y, V_t) \rangle &= \langle y, g_n(t, y + y_t^m, V_t) - g_n(t, y_t^m, V_t) \rangle \\ &\quad + \langle y, g_n(t, y_t^m, V_t) - g_m(t, y_t^m, V_t) \rangle \\ &\leq \kappa(|y|^2) + |y| |g_n(t, y_t^m, V_t) - g_m(t, y_t^m, V_t)|. \end{aligned}$$

Then the assumption (A) is satisfied for the generator $\hat{g}^{n,m}(t, y, V_t)$ of BSDE (3.17) with $\psi(u) = \kappa(u)$, $\lambda = 0$ and $\varphi_t = |g_n(t, y_t^m, V_t) - g_m(t, y_t^m, V_t)|$. Thus, it follows from Proposition 3.1 with $u = 0$ and $\theta = 1$ that there exists a constant $c > 0$ depending only on T such that for $t \in [0, T]$,

$$\begin{aligned} &\mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \\ &\leq c \int_t^T \kappa \left(\mathbf{E} \left[\sup_{r \in [s, T]} |\hat{y}_r^{n,m}|^2 \right] \right) ds + c \mathbf{E} \left[\int_0^T |g_n(s, y_s^m, V_s) - g_m(s, y_s^m, V_s)|^2 ds \right]. \end{aligned} \quad (3.18)$$

On the other hand, it follows from (3.10) that, for each $n, m \geq 1$, $s \in [0, T]$ and each $y \in \mathbb{R}^k$, $dP \times dt$ -a.e.,

$$|g_n(s, y_s^m, V_s) - g_m(s, y_s^m, V_s)| \leq \int_{\{u: |u| \leq 1\}} f^{n,m,s}(u) \psi(u) du$$

with $f^{n,m,s}(u) = |g(s, y_s^m - u/n, V_s) - g(s, y_s^m - u/m, V_s)|$. By (H2), (3.15) and the uniformly continuous function theorem, we can conclude that $dP \times ds$ -a.e., for each $u \in \mathbb{R}^k$, $f^{n,m,s}(u) \rightarrow 0$ as $n, m \rightarrow \infty$. And, it follows from (3.11) and (3.15) that for each $s \in [0, T]$, $n, m \geq 1$ and $u \in \mathbb{R}^k$ with $|u| \leq 1$, $|f^{n,m,s}(u)| \leq 2(K + \mu K + \varphi(r+1))$ and then $|g_n(s, y_s^m, V_s) - g_m(s, y_s^m, V_s)| \leq 2(K + \mu K + \varphi(r+1))$. Thus, applying Lebesgue's dominated convergence theorem twice yields that

$$\lim_{n, m \rightarrow \infty} \mathbf{E} \left[\int_0^T |g_n(s, y_s^m, V_s) - g_m(s, y_s^m, V_s)|^2 ds \right] = 0. \quad (3.19)$$

Now, for each $t \in [0, T]$, we set

$$h^{n,m}(t) := \mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \quad \text{and} \quad h(t) = \limsup_{n, m \rightarrow \infty} h^{n,m}(t).$$

Then, $h(t)$ is well defined by (3.15). By taking the $\limsup$ in (3.18), in view of Fatou's lemma, the monotonicity and continuity of $\kappa(\cdot)$ and (3.19), we can deduce that for each $t \in [0, T]$, $h(t) \leq c \int_t^T \kappa(h(s)) ds$. Thus, in view of $\int_0^+ \kappa^{-1}(u) du = +\infty$, Bihari's inequality yields that

$$\lim_{n, m \rightarrow \infty} \left\{ \mathbf{E} \left[\sup_{r \in [0, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_0^T |\hat{z}_s^{n,m}|^2 ds \right] \right\} = 0,$$

which means that $\{(y_t^n, z_t^n)_{t \in [0, T]}\}_{n=1}^\infty$ is indeed a Cauchy sequence in the process space $\mathcal{S}^2(0, T; \mathbb{R}^k) \times M^2(0, T; \mathbb{R}^{k \times d})$.

Finally, let $(y_t, z_t)_{t \in [0, T]}$ be the limit process of the sequence $\{(y_t^n, z_t^n)_{t \in [0, T]}\}_{n=1}^\infty$ in $\mathcal{S}^2(0, T; \mathbb{R}^k) \times M^2(0, T; \mathbb{R}^{k \times d})$. We pass to the limit in uniform convergence in probability (ucp for short) for BSDE (3.16), in view of (3.10)–(3.12), (3.15), (H2) and Lebesgue's dominated convergence theorem, to see that (y, z) solves BSDE (3.8).

Step 2 In this step, we will eliminate the bounded condition with respect to the process $(V_t)_{t \in [0, T]}$ in Step 1. That is to say, we shall prove that for each $V \in M^2(0, T; \mathbb{R}^{k \times d})$ and $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^k)$, if there exists a positive constant K such that

$$dP\text{-a.s.}, |\xi| \leq K, \text{ and } dP \times dt\text{-a.e.}, |g(t, 0, 0)| \leq K, \quad (3.20)$$

then there exists a unique solution to BSDE (3.8).

For each $n \geq 1$ and $z \in \mathbb{R}^{k \times d}$, denote $q_n(z) = zn/(|z| \vee n)$. Then $|q_n(z)| \leq |z| \wedge n$. It follows from Step 1 that for each $n \geq 1$, there exists a solution $(y_t^n, z_t^n)_{t \in [0, T]}$ to the following BSDE:

$$y_t^n = \xi + \int_t^T g(s, y_s^n, q_n(V_s)) ds - \int_t^T z_s^n dB_s, \quad t \in [0, T]. \quad (3.21)$$

In the sequel, we shall show that $\{(y_t^n, z_t^n)_{t \in [0, T]}\}_{n=1}^\infty$ is a Cauchy sequence in the process space $\mathcal{S}^2(0, T; \mathbb{R}^k) \times M^2(0, T; \mathbb{R}^{k \times d})$. In fact, for each $n \geq 1$ and $m \geq 1$, let $\hat{y}^{n,m} = y^n - y^m$, $\hat{z}^{n,m} = z^n - z^m$. Then

$$\hat{y}_t^{n,m} = \int_t^T \hat{g}^{n,m}(s, \hat{y}_s^{n,m}, V_s) ds - \int_t^T \hat{z}_s^{n,m} dB_s, \quad t \in [0, T], \quad (3.22)$$

where $\hat{g}^{n,m}(s, y, V_s) = g(s, y + y_s^m, q_n(V_s)) - g(s, y_s^m, q_m(V_s))$ for each $y \in \mathbb{R}^k$. It follows from (H1) and (H4) that for each $y \in \mathbb{R}^k$, $dP \times dt$ -a.e.,

$$\begin{aligned} \langle y, \hat{g}^{n,m}(t, y, V_t) \rangle &= \langle y, g(t, y + y_t^m, q_n(V_t)) - g(t, y_t^m, q_n(V_t)) \rangle \\ &\quad + \langle y, g(t, y_t^m, q_n(V_t)) - g(t, y_t^m, q_m(V_t)) \rangle \\ &\leq \kappa(|y|^2) + \mu|y||q_n(V_t) - q_m(V_t)|. \end{aligned}$$

Then the assumption (A) is satisfied for the generator $\hat{g}^{n,m}(t, y, V_t)$ of BSDE (3.22) with $\psi(u) = \kappa(u)$, $\lambda = 0$ and $\varphi_t = \mu|q_n(V_t) - q_m(V_t)|$. Thus, it follows from Proposition 3.1 with $u = 0$ and $\theta = 1$ that there exists a constant $c > 0$ depending only on T such that for $t \in [0, T]$,

$$\begin{aligned} &\mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \\ &\leq c \int_t^T \kappa \left(\mathbf{E} \left[\sup_{r \in [s, T]} |\hat{y}_r^{n,m}|^2 \right] \right) ds + c\mu^2 \mathbf{E} \left[\int_0^T |q_n(V_s) - q_m(V_s)|^2 ds \right]. \end{aligned} \quad (3.23)$$

Since $\kappa(x) \leq A(x+1)$ for each $x \geq 0$, applying Gronwall's inequality yields that for each $t \in [0, T]$ and each $n, m \geq 1$,

$$\mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \leq \left(cAT + 2c\mu^2 \mathbf{E} \left[\int_0^T |V_s|^2 ds \right] \right) \cdot e^{cAT}.$$

Thus, by taking the limsup in (3.23) with respect to n, m and using Fatou's lemma, the monotonicity and continuity of $\kappa(\cdot)$ and Bihari's inequality, we know that $\{(y_t^n, z_t^n)_{t \in [0, T]}\}_{n=1}^\infty$ is indeed a Cauchy sequence in $\mathcal{S}^2(0, T; \mathbb{R}^k) \times M^2(0, T; \mathbb{R}^{k \times d})$.

Finally, let $(y_t, z_t)_{t \in [0, T]}$ be the limit process of the sequence $\{(y_t^n, z_t^n)_{t \in [0, T]}\}_{n=1}^\infty$ in $\mathcal{S}^2(0, T; \mathbb{R}^k) \times M^2(0, T; \mathbb{R}^{k \times d})$. We pass to the limit in ucp for BSDE (3.21), in view of (H2)–(H4) and Lebesgue's dominated convergence theorem, to see that $(y_t, z_t)_{t \in [0, T]}$ solves BSDE (3.8).

Step 3 We shall prove that for each $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^k)$, if (3.20) holds, then there exists a unique solution to the BSDE with parameters (ξ, T, g) .

By Step 2, we can construct the Picard's iterative sequence. Let us set as usual $(y_t^0, z_t^0) = (0, 0)$ and define recursively, for each $n \geq 1$,

$$y_t^n = \xi + \int_t^T g(s, y_s^n, z_s^{n-1}) ds - \int_t^T z_s^n dB_s, \quad t \in [0, T]. \quad (3.24)$$

It follows from (H1) and (H4) that for each $y \in \mathbb{R}^k$, $dP \times dt$ -a.e.,

$$\begin{aligned} \langle y, g(t, y, z_t^{n-1}) \rangle &= \langle y, g(t, y, z_t^{n-1}) - g(t, 0, z_t^{n-1}) \rangle + |y| |g(t, 0, z_t^{n-1})| \\ &\leq \kappa(|y|^2) + |y|(|g(t, 0, 0)| + \mu|z_t^{n-1}|). \end{aligned}$$

Then the assumption (A) is satisfied for the generator $g(t, y, z_t^{n-1})$ of BSDE (3.24) with $\psi(u) = \kappa(u)$, $\lambda = 0$ and $\varphi_t \equiv |g(t, 0, 0)| + \mu|z_t^{n-1}|$. Thus, it follows from Proposition 3.1 with $\theta = 16\mu^2$ and $u = 0$ that there exists a constant $c > 0$ depending only on μ such that for $t \in [0, T]$,

$$\begin{aligned} &\mathbf{E} \left[\sup_{r \in [t, T]} |y_r^n|^2 \right] + \mathbf{E} \left[\int_t^T |z_s^n|^2 ds \right] \\ &\leq e^{c(T-t)} M(t) + ce^{c(T-t)} \int_t^T \kappa \mathbf{E} \left[\sup_{r \in [s, T]} |y_r^n|^2 \right] ds + \frac{e^{c(T-t)}}{8} \mathbf{E} \left[\int_t^T |z_s^{n-1}|^2 ds \right], \end{aligned}$$

where, in view of (3.20),

$$M(t) = c\mathbf{E}[|\xi|^2] + \frac{1}{8\mu^2} \mathbf{E} \left[\int_t^T |g(s, 0, 0)|^2 ds \right] \leq cK^2 + \frac{K^2 T}{8\mu^2} =: M.$$

Let us set $T_1 = \max\{T - \ln 2/c, T - \ln 2/(2cA), 0\}$. Then for each $t \in [T_1, T]$, we have $e^{c(T-t)} \leq 2$, $e^{2cA(T-t)} \leq 2$ and

$$\begin{aligned} &\mathbf{E} \left[\sup_{r \in [t, T]} |y_r^n|^2 \right] + \mathbf{E} \left[\int_t^T |z_s^n|^2 ds \right] \\ &\leq 2M + 2c \int_t^T \kappa \left(\mathbf{E} \left[\sup_{r \in [s, T]} |y_r^n|^2 \right] \right) ds + \frac{1}{4} \mathbf{E} \left[\int_t^T |z_s^{n-1}|^2 ds \right]. \end{aligned}$$

Furthermore, note that $\kappa(x) \leq A(x+1)$ for each $x \geq 0$. Gronwall's inequality yields that for each $t \in [T_1, T]$,

$$\begin{aligned} \mathbf{E} \left[\sup_{r \in [t, T]} |y_r^n|^2 \right] + \mathbf{E} \left[\int_t^T |z_s^n|^2 ds \right] &\leq \left(2M + 2cAT + \frac{1}{4} \mathbf{E} \left[\int_t^T |z_s^{n-1}|^2 ds \right] \right) \cdot e^{2cA(T-t)} \\ &\leq 4M + 4cAT + \frac{1}{2} \mathbf{E} \left[\int_t^T |z_s^{n-1}|^2 ds \right], \end{aligned}$$

from which we can deduce by induction that for each $n \geq 1$ and $t \in [T_1, T]$,

$$\mathbf{E} \left[\sup_{r \in [t, T]} |y_r^n|^2 \right] + \mathbf{E} \left[\int_t^T |z_s^n|^2 ds \right] \leq 8M + 8cAT. \quad (3.25)$$

In the sequel, for each $n \geq 1$ and $m \geq 1$, let $\hat{y}^{n,m} = y^n - y^m$, $\hat{z}^{n,m} = z^n - z^m$. Then

$$\hat{y}_t^{n,m} = \int_t^T \hat{g}^{n,m}(s, \hat{y}_s^{n,m}) ds - \int_t^T \hat{z}_s^{n,m} dB_s, \quad t \in [0, T], \quad (3.26)$$

where $\hat{g}^{n,m}(s, y) = g(s, y + y_s^m, z_s^{n-1}) - g(s, y_s^m, z_s^{m-1})$ for each $y \in \mathbb{R}^k$. It follows from (H1) and (H4) that for each $y \in \mathbb{R}^k$, $dP \times dt$ -a.e.,

$$\langle y, \hat{g}^{n,m}(t, y) \rangle \leq \kappa(|y|^2) + \mu|y||z_t^{n-1} - z_t^{m-1}| = \kappa(|y|^2) + \mu|y||\hat{z}_t^{n-1, m-1}|.$$

Then the assumption (A) is satisfied for the generator $\hat{g}^{n,m}(t, y)$ of BSDE (3.26) with $\psi(u) = \kappa(u)$, $\lambda = 0$ and $\varphi_t \equiv \mu|\hat{z}_t^{n-1, m-1}|$. Thus, it follows from Proposition 3.1 with $\theta = 8\mu^2$ and $u = 0$ that there exists a constant $\bar{c} > 0$ depending only on μ such that for $t \in [0, T]$,

$$\begin{aligned} & \mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \\ & \leq \bar{c} e^{\bar{c}(T-t)} \int_t^T \kappa \left(\mathbf{E} \left[\sup_{r \in [s, T]} |\hat{y}_r^{n,m}|^2 \right] \right) ds + \frac{e^{\bar{c}(T-t)}}{8} \mathbf{E} \left[\int_t^T |\hat{z}_s^{n-1, m-1}|^2 ds \right]. \end{aligned}$$

Without loss of generality, we can assume that $\bar{c} = c$. Let us recall $T_1 = \max\{T - \ln 2/c, T - \ln 2/(2cA), 0\}$. Then for each $t \in [T_1, T]$, we have

$$\begin{aligned} & \mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \\ & \leq 2c \int_t^T \kappa \left(\mathbf{E} \left[\sup_{r \in [s, T]} |\hat{y}_r^{n,m}|^2 \right] \right) ds + \frac{1}{4} \mathbf{E} \left[\int_t^T |\hat{z}_s^{n-1, m-1}|^2 ds \right]. \end{aligned}$$

Now, in view of (3.25), by taking the limsup in the above inequality and using Fatou's lemma, the monotonicity and continuity of $\kappa(\cdot)$ and Bihari's inequality, we know that $\{(y_t^n, z_t^n)_{t \in [T_1, T]}\}_{n=1}^\infty$ is a Cauchy sequence in the process space $\mathcal{S}^2(T_1, T; \mathbb{R}^k) \times M^2(T_1, T; \mathbb{R}^{k \times d})$. Let $(y_t, z_t)_{t \in [T_1, T]}$ be the limit process of this sequence. We pass to the limit in ucp for BSDE (3.24), in view of (H2)–(H4) and Lebesgue's dominated convergence theorem, to see that $(y_t, z_t)_{t \in [T_1, T]}$ is a solution to the BSDE with parameters (ξ, T, g) on $[T_1, T]$. Note that $T - T_1$ is a positive number and depends only on μ and A . We can repeat the above procedure in finite steps to obtain a solution to the BSDE with parameters (ξ, T, g) on $[T_2, T_1]$, $[T_3, T_2]$, $\dots$, and then on $[0, T]$.

Step 4 We shall prove that, for each $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^k)$, there exists a unique solution to the BSDE with parameters (ξ, T, g) .

We approximate $g(t, y, z)$, which satisfies the bounds assumed in Step 3 by data as follows (see (3.20)). For each $n \geq 1$, define $q_n(x) = xn/(|x| \vee n)$ for each $x \in \mathbb{R}^k$, and let

$$\xi_n := q_n(\xi) \quad \text{and} \quad g_n(t, y, z) := g(t, y, z) - g(t, 0, 0) + q_n(g(t, 0, 0)). \quad (3.27)$$

Clearly, the g_n satisfies the assumptions of Step 3,

$$\mathbf{E} [|\xi_n - \xi|^2] \rightarrow 0 \quad \text{and} \quad \mathbf{E} \left[\int_0^T |q_n(g(s, 0, 0)) - g(s, 0, 0)|^2 ds \right] \rightarrow 0 \quad (3.28)$$

as $n \rightarrow \infty$ by (H5). For each $n \geq 1$, let $(y_t^n, z_t^n)_{t \in [0, T]}$ denote the unique solution to the following BSDE:

$$y_t^n = \xi_n + \int_t^T g_n(s, y_s^n, z_s^n) ds - \int_t^T z_s^n dB_s, \quad t \in [0, T]. \quad (3.29)$$

In the sequel, for each $n \geq 1$ and $m \geq 1$, let $\hat{y}^{n,m} = y^n - y^m$, $\hat{z}^{n,m} = z^n - z^m$. Then

$$\hat{y}_t^{n,m} = \xi_n - \xi_m + \int_t^T \hat{g}^{n,m}(s, \hat{y}_s^{n,m}, \hat{z}_s^{n,m}) ds - \int_t^T \hat{z}_s^{n,m} dB_s, \quad t \in [0, T], \quad (3.30)$$

where $\hat{g}^{n,m}(s, y, z) := g_n(s, y + y_s^m, z + z_s^m) - g_m(s, y_s^m, z_s^m)$ for each $(y, z) \in \mathbb{R}^k \times \mathbb{R}^{k \times d}$. In view of (3.27), it follows from (H1) and (H4) that $dP \times dt$ -a.e., for each $(y, z) \in \mathbb{R}^k \times \mathbb{R}^{k \times d}$,

$$\begin{aligned} \langle y, \hat{g}^{n,m}(t, y, z) \rangle &= \langle y, g_n(t, y + y_t^m, z + z_t^m) - g_m(t, y_t^m, z_t^m) \rangle \\ &\quad + \langle y, g_m(t, y + y_t^m, z + z_t^m) - g_m(t, y_t^m, z_t^m) \rangle \\ &= \langle y, q_n(g(t, 0, 0)) - q_m(g(t, 0, 0)) \rangle \\ &\quad + \langle y, g(t, y + y_t^m, z + z_t^m) - g(t, y_t^m, z_t^m) \rangle \\ &\leq |y| |q_n(g(t, 0, 0)) - q_m(g(t, 0, 0))| + \kappa(|y|^2) + \mu|y||z|. \end{aligned}$$

Then the assumption (A) is satisfied for the generator $\hat{g}^{n,m}(t, y, z)$ of BSDE (3.30) with $\psi(u) = \kappa(u)$, $\lambda = \mu$ and $\varphi_t = |q_n(g(t, 0, 0)) - q_m(g(t, 0, 0))|$. Thus, it follows from Proposition 3.1 with $u = 0$ and $\theta = 1$ that there exists a constant $c > 0$ depending only on T and μ such that for $t \in [0, T]$,

$$\begin{aligned} &\mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \\ &\leq c \mathbf{E}[|\xi_n - \xi_m|^2] + c \int_t^T \kappa \left(\mathbf{E} \left[\sup_{r \in [s, T]} |\hat{y}_r^{n,m}|^2 \right] \right) ds \\ &\quad + c \mathbf{E} \left[\int_0^T |q_n(g(s, 0, 0)) - q_m(g(s, 0, 0))|^2 ds \right]. \end{aligned} \quad (3.31)$$

Since $\kappa(x) \leq A(x + 1)$ for each $x \geq 0$, applying Gronwall's inequality yields that for each $t \in [0, T]$ and each $n, m \geq 1$,

$$\begin{aligned} &\mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^{n,m}|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^{n,m}|^2 ds \right] \\ &\leq \left(2c \mathbf{E}[|\xi|^2] + cAT + 2c \mathbf{E} \left[\int_0^T |g(s, 0, 0)|^2 ds \right] \right) \cdot e^{cAT}. \end{aligned}$$

Thus, in view of (3.28), by taking the limsup in (3.31) with respect to n, m and using Fatou's lemma, the monotonicity and continuity of $\kappa(\cdot)$ and Bihari's inequality we know that $\{(y_t^n, z_t^n)_{t \in [0, T]}\}_{n=1}^\infty$ is a Cauchy sequence in the process space $\mathcal{S}^2(0, T; \mathbb{R}^k) \times \mathcal{M}^2(0, T; \mathbb{R}^{k \times d})$. Let $(y_t, z_t)_{t \in [0, T]}$ be the limit process of the sequence $\{(y_t^n, z_t^n)_{t \in [0, T]}\}_{n=1}^\infty$. We pass to the limit in ucp for BSDE (3.29), in view of (H2)–(H4) and Lebesgue's dominated convergence theorem, to see that $(y_t, z_t)_{t \in [0, T]}$ solves the BSDE with parameters (ξ, T, g) . Thus, we prove the existence part and finally complete the proof of Theorem 2.1. $\square$

4 A Stability Theorem

In this section, we shall prove a stability theorem for solutions to BSDEs with generators satisfying (H1) and (H4).

In the sequel, for each $\varepsilon \geq 0$, let $(y_t^\varepsilon, z_t^\varepsilon)_{t \in [0, T]}$ be a solution of the following BSDE depending

on ε :

$$y_t^\varepsilon = \xi^\varepsilon + \int_t^T g^\varepsilon(s, y_s^\varepsilon, z_s^\varepsilon) ds - \int_t^T z_s^\varepsilon dB_s, \quad t \in [0, T].$$

Furthermore, we introduce the following assumptions:

(A1) $\xi^\varepsilon \in L^2(\mathcal{F}_T; \mathbb{R}^k)$ for $\varepsilon \geq 0$ and all of g^ε satisfy assumptions (H1) and (H4) with the same $\kappa(\cdot)$ and μ .

(A2) $\mathbf{E}[|\xi^\varepsilon - \xi^0|^2 + \int_0^T |g^\varepsilon(s, y_s^0, z_s^0) - g^0(s, y_s^0, z_s^0)|^2 ds] \rightarrow 0$ as $\varepsilon \rightarrow 0$.

The following Theorem 4.1 is the main result of this section.

Theorem 4.1 Under assumptions (A1) and (A2), we have

$$\mathbf{E} \left[\sup_{r \in [0, T]} |y_r^\varepsilon - y_r^0|^2 \right] + \mathbf{E} \left[\int_0^T |z_s^\varepsilon - z_s^0|^2 ds \right] \rightarrow 0, \quad \text{as } \varepsilon \rightarrow 0. \quad (4.1)$$

Proof Let $\hat{y}^\varepsilon = y^\varepsilon - y^0$, $\hat{z}^\varepsilon = z^\varepsilon - z^0$ and $\hat{\xi}^\varepsilon = \xi^\varepsilon - \xi^0$. Then

$$\hat{y}_t^\varepsilon = \hat{\xi}^\varepsilon + \int_t^T \hat{g}^\varepsilon(s, \hat{y}_s^\varepsilon, \hat{z}_s^\varepsilon) ds - \int_t^T \hat{z}_s^\varepsilon dB_s, \quad t \in [0, T], \quad (4.2)$$

where $\hat{g}^\varepsilon(s, y, z) := g^\varepsilon(s, y + y_s^0, z + z_s^0) - g^0(s, y_s^0, z_s^0)$ for each $(y, z) \in \mathbb{R}^k \times \mathbb{R}^{k \times d}$. It follows from (A1) that $dP \times dt$ -a.e., for each $(y, z) \in \mathbb{R}^k \times \mathbb{R}^{k \times d}$ and $\varepsilon \geq 0$,

$$\begin{aligned} \langle y, \hat{g}^\varepsilon(t, y, z) \rangle &= \langle y, g^\varepsilon(t, y + y_t^0, z + z_t^0) - g^\varepsilon(t, y_t^0, z_t^0) \rangle \\ &\quad + \langle y, g^\varepsilon(t, y_t^0, z_t^0) - g^0(t, y_t^0, z_t^0) \rangle \\ &\leq \kappa(|y|^2) + \mu|y||z| + |y||g^\varepsilon(t, y_t^0, z_t^0) - g^0(t, y_t^0, z_t^0)|. \end{aligned}$$

Then the assumption (A) is satisfied for the generator $\hat{g}^\varepsilon(t, y, z)$ of BSDE (4.2) with $\psi(u) = \kappa(u)$, $\lambda = \mu$ and $\varphi_t = |g^\varepsilon(t, y_t^0, z_t^0) - g^0(t, y_t^0, z_t^0)|$. Thus, it follows from Proposition 3.1 with $u = 0$ and $\theta = 1$ that there exists a constant $c > 0$ depending only on μ and T such that for $t \in [0, T]$,

$$\begin{aligned} \mathbf{E} \left[\sup_{r \in [t, T]} |\hat{y}_r^\varepsilon|^2 \right] + \mathbf{E} \left[\int_t^T |\hat{z}_s^\varepsilon|^2 ds \right] &\leq c \mathbf{E}[|\hat{\xi}^\varepsilon|^2] + c \int_t^T \kappa \left(\mathbf{E} \left[\sup_{r \in [s, T]} |\hat{y}_r^\varepsilon|^2 \right] \right) ds \\ &\quad + c \mathbf{E} \left[\int_0^T |g^\varepsilon(s, y_s^0, z_s^0) - g^0(s, y_s^0, z_s^0)|^2 ds \right]. \end{aligned} \quad (4.3)$$

Furthermore, in view of (A2) and the fact that $\kappa(\cdot)$ is of linear growth, Gronwall's inequality yields that there exists a constant $M > 0$ independent of ε such that

$$\mathbf{E} \left[\sup_{r \in [0, T]} |\hat{y}_r^\varepsilon|^2 \right] + \mathbf{E} \left[\int_0^T |\hat{z}_s^\varepsilon|^2 ds \right] \leq M.$$

Then, in view of (A2), by taking the limsup in (4.3) and using Fatou's lemma, the monotonicity and continuity of $\kappa(\cdot)$ and Bihari's inequality, we obtain (4.1). The proof of Theorem 4.1 is then complete. $\square$

Remark 4.2 It is easy to see that the uniqueness of Theorem 2.1 is an immediate corollary of Theorem 4.1.

5 A Comparison Theorem

In this section, we restrict ourselves to the case $k = 1$ and prove the following comparison theorem for solutions to BSDEs with generators satisfying (H1) and (H4). This theorem improves the comparison results in [5] and [17].

Theorem 5.1 Let $\xi, \xi' \in L^2(\mathcal{F}_T; \mathbb{R}^k)$, g and g' be two generators of BSDEs, and (y, z) (resp. (y', z')) be a solution to the BSDE with parameters (ξ, T, g) (resp. (ξ', T, g')). If dP -a.s., $\xi \leq \xi'$, g satisfies (H1) and (H4) and $dP \times dt$ -a.e., $g(t, y'_t, z'_t) \leq g'(t, y'_t, z'_t)$ (or g' satisfies (H1) and (H4) and $dP \times dt$ -a.e., $g(t, y_t, z_t) \leq g'(t, y_t, z_t)$), then for each $t \in [0, T]$, we have

$$dP\text{-a.s.}, y_t \leq y'_t.$$

Proof The main idea of the proof is motivated by [5]. We first assume that dP -a.s., $\xi \leq \xi'$, g satisfies (H1) and (H4) and $dP \times dt$ -a.e., $g(t, y'_t, z'_t) \leq g'(t, y'_t, z'_t)$. Setting $\hat{y}_t = y_t - y'_t$, $\hat{z}_t = z_t - z'_t$, we have

$$\hat{y}_t = (\xi - \xi') + \int_t^T [g(s, y_s, z_s) - g'(s, y'_s, z'_s)] ds - \int_t^T \hat{z}_s dB_s.$$

It follows from Lemma 2.2 in [5] that

$$\begin{aligned} (\hat{y}_t^+)^2 + \int_t^T 1_{\hat{y}_s > 0} |\hat{z}_s|^2 ds &= [(\xi - \xi')^+]^2 - 2 \int_t^T \hat{y}_s^+ \hat{z}_s dB_s \\ &\quad + 2 \int_t^T \hat{y}_s^+ [g(s, y_s, z_s) - g'(s, y'_s, z'_s)] ds. \end{aligned} \quad (5.1)$$

Since $g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)$ is non-positive, we have

$$\begin{aligned} g(s, y_s, z_s) - g'(s, y'_s, z'_s) &= g(s, y_s, z_s) - g(s, y'_s, z'_s) + g(s, y'_s, z'_s) - g'(s, y'_s, z'_s) \\ &\leq g(s, y_s, z_s) - g(s, y'_s, z'_s) + g(s, y'_s, z'_s) - g'(s, y'_s, z'_s); \end{aligned}$$

and using (H1) and (H4) for g , we deduce that

$$2\hat{y}_s^+ [g(s, y_s, z_s) - g'(s, y'_s, z'_s)] \leq 2\kappa((\hat{y}_s^+)^2) + (\hat{y}_s^+)^2 \mu^2 + 1_{\hat{y}_s > 0} |\hat{z}_s|^2. \quad (5.2)$$

Thus, in view of $\xi \leq \xi'$, it follows from (5.1) and (5.2) that for each $t \in [0, T]$,

$$(\hat{y}_t^+)^2 \leq \int_t^T \bar{\kappa}((\hat{y}_s^+)^2) ds - \int_t^T \hat{y}_s^+ \hat{z}_s dB_s,$$

where $\bar{\kappa}(u) = 2\kappa(u) + \mu^2 u$ is again a nondecreasing and concave function with $\bar{\kappa}(0) = 0$ and $\bar{\kappa}(u) > 0$ for $u > 0$. Note that $(\int_0^t \hat{y}_s^+ \hat{z}_s dB_s)_{t \in [0, T]}$ is a martingale by BDG's inequality. It follows from Jensen's inequality that for each $t \in [0, T]$,

$$\mathbf{E}[(\hat{y}_t^+)^2] \leq \int_t^T \bar{\kappa}(\mathbf{E}[(\hat{y}_s^+)^2]) ds. \quad (5.3)$$

Furthermore, since $\kappa(\cdot)$ is a concave function and $\kappa(0) = 0$, we have $\kappa(u) \geq \kappa(1)u$ for each $u \in [0, 1]$, and then for each $0 \leq u \leq 1$,

$$\frac{1}{\bar{\kappa}(u)} = \frac{1}{2\kappa(u) + \mu^2 u} \geq \frac{1}{2\kappa(u) + \frac{\mu^2}{\kappa(1)}\kappa(u)} = \frac{\kappa(1)}{2\kappa(1) + \mu^2} \cdot \frac{1}{\kappa(u)}.$$

As a result, $\int_{0+} \bar{\kappa}^{-1}(u) du = +\infty$. Thus, in view of (5.3), Bihari's inequality yields that for each $t \in [0, T]$, $\mathbf{E}[(\hat{y}_t^+)^2] = 0$ and then dP -a.s., $y_t \leq y'_t$.

Now, let us assume that dP -a.s., $\xi \leq \xi'$, g' satisfies (H1) and (H4) and $dP \times dt$ -a.e., $g(t, y_t, z_t) \leq g'(t, y_t, z_t)$. Since $g(s, y_s, z_s) - g'(s, y_s, z_s)$ is non-positive, we have

$$g(s, y_s, z_s) - g'(s, y'_s, z'_s) = g(s, y_s, z_s) - g'(s, y_s, z_s) + g'(s, y_s, z_s) - g'(s, y'_s, z'_s)$$

$$\leq g'(s, y_s, z_s) - g'(s, y'_s, z_s) + g'(s, y'_s, z_s) - g'(s, y'_s, z'_s).$$

Furthermore, using (H1) and (H4) for g' , we know that the inequality (5.2) holds still true. Therefore, the same arguments as above yield that for each $t \in [0, T]$, dP -a.s., $y_t \leq y'_t$. The theorem is proved. $\square$

From Theorem 5.1, the following corollary is immediate.

Corollary 5.2 *Assume that one of g and g' satisfies (H1) and (H4). Let $\xi, \xi' \in L^2(\mathcal{F}_T, \mathbb{R}^1)$, and let (y, z) (resp. (y', z')) be a solution to the BSDE with parameters (ξ, T, g) (resp. (ξ', T, g')). If dP -a.s., $\xi \leq \xi'$ and $dP \times dt$ -a.e., $\forall y, z$, $g(t, y, z) \leq g'(t, y, z)$, then for each $t \in [0, T]$, dP -a.s., $y_t \leq y'_t$.*

Remark 5.3 It follows from Remark 2.2 that two classical comparison theorems obtained by [5] and [17] have been generalized by Theorem 5.1 and Corollary 5.2.

6 Appendix

Some technical details in Section 2 are put in this section. Let us start with the following lemma, which will be frequently used later.

Lemma 6.1 *Let $f(\cdot)$ be a nondecreasing continuous function on $\mathbb{R}^+$ with $f(0) = 0$. Then, the following two statements hold true:*

- (i) *If $f(\cdot)$ is concave on $\mathbb{R}^+$, then $f(x)/x$ ($x > 0$) is a non-increasing function.*
- (ii) *If $f(x)/x$ ($x > 0$) is a non-increasing function, then there exists a nondecreasing and concave function $p(\cdot)$ defined on $\mathbb{R}^+$ such that $f(x) \leq p(x) \leq 2f(x)$ for each $x \geq 0$.*

Proof We first prove (i). Let $f(\cdot)$ be concave on $\mathbb{R}^+$ with $f(0) = 0$. For each $x \in \mathbb{R}^+$ and $\alpha \in [0, 1]$, we have

$$f(\alpha x) = f(\alpha x + (1 - \alpha) \cdot 0) \geq \alpha f(x) + (1 - \alpha)f(0) = \alpha f(x).$$

Therefore, for each $x > 0$ and $y > 0$, if $x \leq y$, then $f(x) = f(yx/y) \geq f(y)x/y$, which means that $f(x)/x$ ($x > 0$) is a non-increasing function.

Let us now turn to the proof of (ii). Assume that $f(x)/x$ ($x > 0$) is a non-increasing function. For each $x > 0$ and $y > 0$, we have

$$\frac{f(x+y)}{x+y} \leq \frac{f(x)}{x} \quad \text{and} \quad \frac{f(x+y)}{x+y} \leq \frac{f(y)}{y},$$

and then

$$\frac{xf(x+y)}{x+y} \leq f(x) \quad \text{and} \quad \frac{yf(x+y)}{x+y} \leq f(y).$$

Thus, for each $x \geq 0$ and $y \geq 0$, one has $f(x+y) \leq f(x) + f(y)$. Consequently, $f(x)$ is a continuous modular function and the result follows by letting $p(x)$ be the least concave majorant of $f(x)$ (see [13, pp. 499–500] for details). $\square$

In the sequel, we will prove that $(H1c) \Rightarrow (H1b) \Rightarrow (H1a) \Rightarrow (H1)$. Similarly, we can prove that $(H1c)' \Rightarrow (H1b)' \Rightarrow (H1a)' \Rightarrow (H1)'$.

First, let us show that $(H1c) \Rightarrow (H1b)$. Assume that g satisfies (H1c) with the non-decreasing

concave function $\rho(\cdot)$. Then we have, $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq \sqrt{\bar{\rho}(|y_1 - y_2|^2)},$$

where $\bar{\rho}(x) = \rho^2(\sqrt{x})$ for each $x \geq 0$. Obviously, $\bar{\rho}(\cdot)$ is a nondecreasing continuous function with $\bar{\rho}(0) = 0$ and $\bar{\rho}(x) > 0$ for $x > 0$. It follows from (i) of Lemma 6.1 that $\bar{\rho}(x)/x = (\rho(\sqrt{x})/\sqrt{x})^2$ is a non-increasing function for $x > 0$. Furthermore, by virtue of (ii) of Lemma 6.1, we know that there exists a nondecreasing concave function $\kappa(\cdot)$ such that $\bar{\rho}(x) \leq \kappa(x) \leq 2\bar{\rho}(x)$ for each $x \geq 0$. Then we have, $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq \sqrt{\kappa(|y_1 - y_2|^2)},$$

and

$$\int_{0+} \frac{du}{\kappa(u)} \geq \frac{1}{2} \int_{0+} \frac{du}{\bar{\rho}(u)} = \frac{1}{2} \int_{0+} \frac{du}{\rho^2(\sqrt{u})} = \int_{0+} \frac{xdx}{\rho^2(x)} = +\infty.$$

Hence (H1b) holds for g .

Then, let us show that (H1b) $\Rightarrow$ (H1a). Assume that g satisfies (H1b) with the non-decreasing concave function $\kappa(\cdot)$. Then we have, $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq \bar{\kappa}(|y_1 - y_2|),$$

where $\bar{\kappa}(x) = \sqrt{\kappa(x^2)}$ for each $x \geq 0$. Obviously, $\bar{\kappa}(\cdot)$ is a nondecreasing continuous function with $\bar{\kappa}(0) = 0$ and $\bar{\kappa}(x) > 0$ for $x > 0$. It follows from (i) of Lemma 6.1 that $\bar{\kappa}(x)/x = \sqrt{\kappa(x^2)}/x^2$ is a non-increasing function for $x > 0$. Furthermore, by virtue of (ii) of Lemma 6.1, we know that there exists a nondecreasing concave function $\rho(\cdot)$ such that $\bar{\kappa}(x) \leq \rho(x) \leq 2\bar{\kappa}(x)$ for each $x \geq 0$. Then we have, $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$\left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \right\rangle \leq \rho(|y_1 - y_2|).$$

Moreover, it follows from (i) of Lemma 6.1 that

$$\liminf_{u \rightarrow 0+} \frac{\frac{1}{\sqrt{\kappa(u^2)}}}{\frac{u}{\kappa(u^2)}} = \liminf_{u \rightarrow 0+} \sqrt{\frac{\kappa(u^2)}{u^2}} \geq \sqrt{\frac{\kappa(1)}{1}} > 0$$

and

$$\int_{0+} \frac{2udu}{\kappa(u^2)} = \int_{0+} \frac{dx}{\kappa(x)} = +\infty,$$

from which we know that

$$\int_{0+} \frac{du}{\rho(u)} \geq \frac{1}{2} \int_{0+} \frac{du}{\bar{\kappa}(u)} = \int_{0+} \frac{du}{\sqrt{\kappa(u^2)}} = +\infty.$$

Hence (H1a) holds for g .

Finally, we show that (H1a) $\Rightarrow$ (H1). Let the generator g satisfy (H1a) with the non-decreasing concave function $\rho(\cdot)$. Then we have, $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad \langle y_1 - y_2, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \rangle \leq \bar{\rho}(|y_1 - y_2|^2),$$

where $\bar{\rho}(x) = \sqrt{x}\rho(\sqrt{x})$ for each $x \geq 0$. Obviously, $\bar{\rho}(\cdot)$ is a nondecreasing continuous function with $\bar{\rho}(0) = 0$ and $\bar{\rho}(x) > 0$ for $x > 0$. It follows from (i) of Lemma 6.1 that $\bar{\rho}(x)/x = \rho(\sqrt{x})/\sqrt{x}$

is a non-increasing function for $x > 0$. Furthermore, by virtue of (ii) of Lemma 6.1, we know that there exists a nondecreasing concave function $\kappa(\cdot)$ such that $\bar{\rho}(x) \leq \kappa(x) \leq 2\bar{\rho}(x)$ for each $x \geq 0$. Then we have, $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad \langle y_1 - y_2, g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \rangle \leq \kappa(|y_1 - y_2|^2),$$

and

$$\int_{0+} \frac{du}{\kappa(u)} \geq \frac{1}{2} \int_{0+} \frac{du}{\bar{\rho}(u)} = \frac{1}{2} \int_{0+} \frac{du}{\sqrt{u}\rho(\sqrt{u})} = \int_{0+} \frac{dx}{\rho(x)} = +\infty.$$

Hence, (H1) holds for g .

Now, we show that the generator g in Example 2.4 satisfies neither (H1b) nor (H1)'. In fact, if there exists a nondecreasing and continuous function $\kappa(\cdot)$ with $\kappa(0) = 0$ and $\kappa(u) > 0$ for $u > 0$ such that (H1b) holds for this generator g , then we have, $dP \times dt$ -a.e.,

$$\forall y_1, y_2, \left\langle \frac{y_1 - y_2}{|y_1 - y_2|} 1_{|y_1 - y_2| \neq 0}, e^{-y_1} - e^{-y_2} + h(|y_1|) - h(|y_2|) \right\rangle \leq \sqrt{\kappa(|y_1 - y_2|^2)}.$$

Thus, for each $u \in [0, \delta^2]$, letting $y_1 = \sqrt{u}$ and $y_2 = 0$ in the above inequality yields that

$$\kappa(u) \geq (e^{-\sqrt{u}} - 1 + h(\sqrt{u}))^2 = \left(e^{-\sqrt{u}} - 1 - \frac{1}{2}\sqrt{u} \ln u \right)^2.$$

Then

$$\int_{0+} \frac{du}{\kappa(u)} \leq \int_{0+} \frac{du}{(e^{-\sqrt{u}} - 1 - \frac{1}{2}\sqrt{u} \ln u)^2} < +\infty.$$

Hence, this generator g does not satisfy (H1b). Furthermore, if there exists a nondecreasing and concave function $\rho(\cdot)$ such that (H1)' holds for g , then we have, $dP \times dt$ -a.e.,

$$\forall y_1, y_2, \quad |y_1 - y_2| |e^{-y_1} - e^{-y_2} + h(|y_1|) - h(|y_2|)| \leq \rho(|y_1 - y_2|^2).$$

Thus, by letting $y_1 = x, y_2 = 0$ with $x < 0$ in the previous inequality and noticing that $\rho(x)$ is of linear growth, we get that there exists a constant $A > 0$ such that

$$\forall x < 0, \quad |x| |e^{-x} - 1 + h(|x|)| \leq \rho(|x|^2) \leq A|x|^2 + A,$$

which leads to the contradiction that e^{-x} is of linear growth on $\mathbb{R}$. Hence, this generator g does not also satisfy (H1)'.

Let us close this appendix by proving Remark 2.6. That is to say, we need to prove that we can use the continuity instead of the concavity of $\rho(\cdot)$ and $\kappa(\cdot)$ in (H1a)', (H1c)' and (H1b)'. The following assumptions with respect to the generator of BSDEs will be used:

(H1a)* There exists a nondecreasing continuous function $\bar{\rho}(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\bar{\rho}(0) = 0$, $\bar{\rho}(u) > 0$ for $u > 0$ and $\int_{0+} \bar{\rho}^{-1}(u) du = +\infty$ such that $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq \bar{\rho}(|y_1 - y_2|).$$

We first show that (H1a)* $\Rightarrow$ (H1a)'. Assume that (H1a)* with $\bar{\rho}(\cdot)$ holds for g . Then, there exists a $\mathcal{P} \subset \Omega \times [0, T]$ with $dP \times dt((\Omega \times [0, T]) \cap \mathcal{P}^c) = 0$ such that for each $(t, \omega) \in \mathcal{P}$, $y_1, y_2 \in \mathbb{R}^k$ and $z \in \mathbb{R}^{k \times d}$, we have

$$|g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq \bar{\rho}(|y_1 - y_2|). \quad (6.1)$$

Now, for each $r \in \mathbb{R}^+$, let

$$F(r) = \sup\{|g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| : |y_1 - y_2| \leq r, (\omega, t, z) \in \mathcal{P} \times \mathbb{R}^{k \times d}\}.$$

It is clear that $F(0) = 0$. It follows from (6.1) that $F(\cdot)$ is well-defined, nondecreasing and $0 \leq F(r) \leq \bar{\rho}(r)$ for each $r \geq 0$. Thus, in view of the continuity of $\bar{\rho}(\cdot)$ and the fact $\bar{\rho}(0) = 0$, we know that $F(\cdot)$ is right-continuous at 0. Furthermore, for $r, s \geq 0$ and $(y_1, y_2) \in \mathbb{R}^k \times \mathbb{R}^k$ with $r \leq |y_1 - y_2| \leq r + s$, it follows from the definition of $F(\cdot)$ that for each $(t, \omega) \in \mathcal{P}$, $y_1, y_2 \in \mathbb{R}^k$ and $z \in \mathbb{R}^{k \times d}$,

$$\begin{aligned} |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| &\leq \left| g(\omega, t, y_1, z) - g\left(\omega, t, y_1 + \frac{r(y_2 - y_1)}{|y_2 - y_1|}, z\right) \right| \\ &\quad + \left| g\left(\omega, t, y_1 + \frac{r(y_2 - y_1)}{|y_2 - y_1|}, z\right) - g(\omega, t, y_2, z) \right| \\ &\leq F(r) + F(s) \end{aligned}$$

so that, the case $|y_1 - y_2| \leq r$ being trivial, we conclude that $F(\cdot)$ is sub-additive. That is, $F(r + s) \leq F(r) + F(s)$ for each $r, s \geq 0$. As a result, $F(\cdot)$ is a continuous modular function, and then there exists a nondecreasing and concave function $\rho_1(\cdot) : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $F(u) \leq \rho_1(u) \leq 2F(u) \leq 2\bar{\rho}(u)$ for each $u \geq 0$ (see [13, pp. 499–500] for details). Thus, it follows from (6.1) and the definition of $F(\cdot)$ that $dP \times dt$ -a.e., for each $(y_1, y_2, z) \in \mathbb{R}^k \times \mathbb{R}^k \times \mathbb{R}^{k \times d}$,

$$|g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq F(|y_1 - y_2|) \leq \rho_1(|y_1 - y_2|) \leq \rho(|y_1 - y_2|),$$

where $\rho(u) := \rho_1(u) + u$. Clearly, $\rho(\cdot)$ is a nondecreasing and concave function with $\rho(0) = 0$ and $\rho(u) > 0$ for $u > 0$. Thus, for completing the proof that g satisfies (H1a)', it suffices to show that $\int_{0+} \rho^{-1}(u) du = +\infty$. Indeed, if $\rho_1(u) > 0$ for each $u > 0$, since $\rho_1(\cdot)$ is concave on $\mathbb{R}^+$ with $\rho_1(0) = 0$, we have $\rho_1(u) \geq u\rho_1(1)$ for each $u \in (0, 1)$, and then

$$\int_{0+} \frac{du}{\rho(u)} = \int_{0+} \frac{du}{\rho_1(u) + u} \geq \frac{\rho_1(1)}{1 + \rho_1(1)} \int_{0+} \frac{du}{\rho_1(u)} \geq \frac{1}{2} \cdot \frac{\rho_1(1)}{1 + \rho_1(1)} \int_{0+} \frac{du}{\bar{\rho}(u)} = +\infty.$$

Otherwise,

$$\int_{0+} \frac{du}{\rho(u)} = \int_{0+} \frac{du}{\rho_1(u) + u} = \int_{0+} \frac{du}{u} = +\infty.$$

Therefore, (H1a)* $\Rightarrow$ (H1a)'.

The same argument as above yields that the concavity condition with respect to $\rho(\cdot)$ in (H1c)' can be replaced with the continuity condition. Thus, to show Remark 2.6, it suffices to prove that the following assumption (H1b)* implies (H1b)'.

(H1b)* There exists a nondecreasing continuous function $\bar{\kappa}(\cdot) : \mathbb{R}^+ \mapsto \mathbb{R}^+$ with $\bar{\kappa}(0) = 0$, $\bar{\kappa}(u) > 0$ for $u > 0$ and $\int_{0+} \bar{\kappa}^{-1}(u) du = +\infty$ such that $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)|^2 \leq \bar{\kappa}(|y_1 - y_2|^2).$$

In fact, assume that (H1b)* with the function $\bar{\kappa}(\cdot)$ holds for g . Then we have, $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$|g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq h_1(|y_1 - y_2|),$$

where $h_1(u) := \sqrt{\bar{\kappa}(u^2)}$. Obviously, $h_1(\cdot)$ is a nondecreasing continuous function with $h_1(0) = 0$ and $h_1(u) > 0$ for $u > 0$, but is not necessarily concave. However, it follows from the proof

of (H1a)* $\Rightarrow$ (H1a)' that if g satisfies the above condition, there exists a nondecreasing concave function $\kappa_1(\cdot) : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $\kappa_1(0) = 0$, $\kappa_1(u) \leq 2h_1(u)$ for $u \geq 0$, and $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq \kappa_1(|y_1 - y_2|).$$

Thus, $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}$,

$$|g(\omega, t, y_1, z) - g(\omega, t, y_2, z)|^2 \leq \kappa_2(|y_1 - y_2|^2),$$

where $\kappa_2(u) := \kappa_1^2(\sqrt{u}) + u$. Furthermore, in view of the concavity of $\kappa_1(\cdot)$, it follows from (i) and (ii) of Lemma 6.1 that there exists a nondecreasing concave function $\kappa(\cdot)$ such that $\kappa_2(u) \leq \kappa(u) \leq 2\kappa_2(u)$ for each $u \geq 0$. Then we have, $dP \times dt$ -a.e.,

$$\forall y_1, y_2 \in \mathbb{R}^k, z \in \mathbb{R}^{k \times d}, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)|^2 \leq \kappa(|y_1 - y_2|^2).$$

Obviously, $\kappa(0) = 0$ and $\kappa(u) > 0$ for $u > 0$. Then, for completing the proof that g satisfies (H1b)', it suffices to show that $\int_{0+} \kappa^{-1}(u) du = +\infty$. Indeed, since $\kappa_1(\cdot)$ is concave on $\mathbb{R}^+$ with $\kappa_1(0) = 0$, we know that for each $u \in [0, 1]$, $\kappa_1(\sqrt{u}) \geq \sqrt{u}\kappa_1(1)$. Thus, if $\kappa_1(u) > 0$ for each $u > 0$, in view of the fact that

$$\forall u \in [0, 1], \quad u \leq \kappa_1^2(\sqrt{u})/\kappa_1^2(1) \quad \text{and} \quad \kappa_1^2(\sqrt{u}) \leq 4h_1^2(\sqrt{u}) = 4\bar{\kappa}(u),$$

we can deduce that, with $c := 1 + \kappa_1^{-2}(1)$,

$$\int_{0+} \frac{du}{\kappa(u)} \geq \frac{1}{2} \int_{0+} \frac{du}{\kappa_2(u)} = \frac{1}{2} \int_{0+} \frac{du}{\kappa_1^2(\sqrt{u}) + u} \geq \frac{1}{2c} \int_{0+} \frac{du}{\kappa_1^2(\sqrt{u})} \geq \frac{1}{8c} \int_{0+} \frac{du}{\bar{\kappa}(u)} = +\infty.$$

Otherwise,

$$\int_{0+} \frac{du}{\kappa(u)} \geq \frac{1}{2} \int_{0+} \frac{du}{\kappa_1^2(\sqrt{u}) + u} = \frac{1}{2} \int_{0+} \frac{du}{u} = +\infty.$$

Thus, (H1b)' holds and then (H1b)* $\Rightarrow$ (H1b)'.

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